

A Survey on RRAM Technology and MOS Integration

Ashton Billings and Jack Walberer

Abstract—Resistive random-access memory (RRAM) is one of three leading candidates—alongside spin-transfer-torque magnetoresistive RAM (STT-MRAM) and phase-change memory (PCM)—to replace embedded flash at logic nodes below 28 nm, where the floating-gate stack no longer scales. This survey reviews the device, its dominant operating mechanism, and how it is integrated into modern CMOS. The metal–insulator–metal (MIM) cell and the standard taxonomy of resistive switching are first introduced. Within RRAM, the industry has converged on the valence-change mechanism (VCM) using HfO_2 or TaO_x as the switching oxide, for reasons related to process compatibility, thermal stability, and switching current that are made explicit in the comparison with PCM and conductive-bridging RAM (CBRAM). The cell is then placed in its production context: a 1-transistor–1-resistor (1T1R) configuration in which a CMOS MOSFET sets the compliance current, addresses the cell, and isolates unselected cells from sneak-path leakage. The select-FET design problem is examined explicitly, with a worked sizing example on a 0.25 μm planar process from the literature, and the constraints are tracked across the planar bulk, FD-SOI, and FinFET architectures that host RRAM in industry today. A dedicated section addresses reliability, retention drift, and disturb mechanisms, and the paper closes with a brief outlook on analog in-memory computing as the most active forward-looking application.

Index Terms—Resistive random-access memory, RRAM, ReRAM, valence-change mechanism, oxygen vacancy, HfO_2 , TaO_x , embedded non-volatile memory, 1T1R, BEOL integration, FinFET, FD-SOI, in-memory computing.

I. INTRODUCTION

FOR more than two decades, embedded flash (eFlash) has served as the default non-volatile memory in microcontrollers, automotive ICs, and IoT system-on-chips. Its physical foundation—a floating-gate or charge-trap MOSFET that retains charge across a tunnel oxide of roughly 7–10 nm—is incompatible with logic scaling: retention specifications fix a minimum oxide thickness that does not shrink with the logic node, and the supporting high-voltage charge pumps add significant mask count and process complexity. By the time the industry reached the 28 nm node, eFlash had become a process integration burden that foundries were no longer willing to carry forward, and TSMC publicly stopped offering eFlash beyond 28 nm in the late 2010s [5]. The resulting gap created the commercial opening for emerging non-volatile memories.

Three technologies—STT-MRAM, RRAM, and PCM—have emerged as the serious contenders for embedded NVM at advanced nodes, with each occupying a different niche [5]. STT-MRAM is the most mature at FinFET-class nodes, with shipping eMRAM macros at TSMC 22ULL/16 nm, Samsung 28FDS, and GlobalFoundries 22FDX. PCM had a major

standalone deployment in Intel–Micron 3D XPoint (Optane) from 2017–2022, and is currently shipping in embedded form at STMicroelectronics in their 28 nm FD-SOI automotive platform [5]. RRAM is most established at planar bulk and FD-SOI nodes, with embedded products at TSMC 40/28/22 nm [7] and at GlobalFoundries via Weebit Nano. Intel has also presented an embedded RRAM cell in the 22FFL FinFET process using a TaO_x switching layer [8]. RRAM is attractive because its cell is a simple two-terminal metal–insulator–metal (MIM) stack of materials already qualified in CMOS fabrication, it is built in the back-end-of-line (BEOL) above the transistors, and its mask adder is small. The cell does not stand alone; in production it is paired with a select MOSFET in a 1-transistor–1-resistor (1T1R) configuration, which makes the technology directly relevant to MOSFET device modeling.

This survey is organized around four questions, mirroring the project presentation. Section II introduces the RRAM cell and the family of resistive switching mechanisms, focusing on the valence-change mechanism (VCM) that dominates today’s commercial cells. Section III addresses why RRAM is being adopted, why VCM specifically rather than competing RRAM mechanisms, and where it fits among the emerging NVMs and the broader memory hierarchy. Section IV treats the integration of the cell into CMOS through the 1T1R architecture and the select-FET sizing problem, with a worked example on a planar process. Section V addresses reliability, retention, and disturb mechanisms. Section VI compares the three MOSFET architectures—planar bulk, FD-SOI, and FinFET—that host RRAM at production nodes. Section VII concludes with an outlook on analog in-memory computing.

II. WHAT IS RRAM?

A. The RRAM Cell

RRAM is a form of non-volatile memory in which information is stored in the resistance of a two-terminal device [1], [2]. A high-resistance state (HRS) is read as logic 0 and a low-resistance state (LRS) is read as logic 1. The transition from HRS to LRS is called SET, and the reverse transition is RESET. Because the storage element is a passive resistor rather than a charge-storing capacitor, RRAM does not suffer from the tunnel-oxide thickness floor that limits flash scaling.

In an array, the resistive switching element (RS) is paired with a non-linear element (NLE) that selects which cell is being addressed. The two canonical choices for the NLE are a two-terminal selector and a CMOS MOSFET, shown in Fig. 1. The MOSFET-based 1T1R cell on the right is the configuration used in essentially every embedded RRAM product shipping today, because the MOSFET simultaneously addresses the cell, sets the compliance current during programming, and blocks leakage from unselected cells during read [2], [3].

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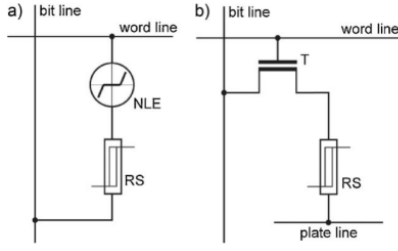


Fig. 1. RRAM cell architectures. (a) Selector-based cell, in which a two-terminal non-linear element (NLE) is placed in series with the resistive switch (RS). (b) 1T1R cell, in which a CMOS MOSFET serves as the select element. Reproduced from [2].

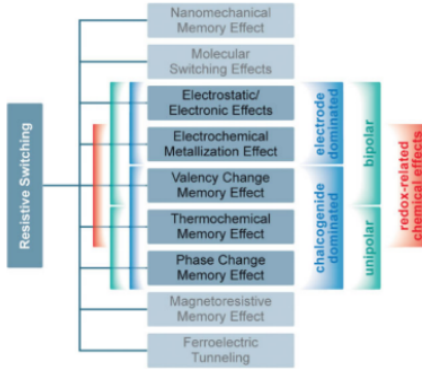


Fig. 2. Taxonomy of resistive switching mechanisms. Reproduced from [2].

B. Resistive Switching Mechanisms

Many physical mechanisms can produce a reversible resistance change in a thin film. Waser *et al.* organize the field into the taxonomy reproduced in Fig. 2, which spans nanomechanical, molecular, electrostatic/electronic, electrochemical metallization, valence-change, thermochemical, phase-change, magnetoresistive, and ferroelectric tunneling effects [2]. The right-hand axis classifies each mechanism by the dominant species (electrode-dominated vs. chalcogenide-dominated, redox-related) and by switching polarity (unipolar vs. bipolar). Of the redox-based mechanisms, three are most relevant to memory.

Electrochemical Metallization Mechanism (ECM). An electrochemically active electrode (typically Ag or Cu) dissolves under positive bias, drifts through a solid electrolyte, and electrocrystallizes into a metallic filament that bridges the electrodes [2]. ECM can operate at low voltages (SET thresholds as low as ~ 0.2 V have been reported in Ag/GeS systems, although typical commercial CBRAM operating windows fall in the 0.5–2 V range), but the metallic filament is sensitive to thermal activation, which complicates retention.

Thermochemical Mechanism (TCM). Switching is driven by Joule heating, with SET and RESET at the same polarity but different current levels (unipolar). NiO is the canonical TCM material [2]. TCM has lost commercial relevance because each RESET is essentially a controlled local meltdown of the filament, and cycle-to-cycle variability is intractable.

Valence-Change Mechanism (VCM). The mobile species is

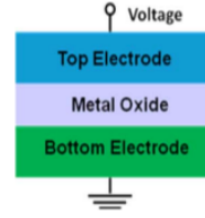


Fig. 3. The metal–insulator–metal (MIM) cell stack. A thin transition-metal oxide is sandwiched between two metal electrodes. Reproduced from [1].

the oxygen anion (or, equivalently, the oxygen vacancy V_O). Under field, oxygen drifts away from a region of the oxide, leaving an oxygen-deficient sub-stoichiometric path of higher conductivity. Because metal cations along the path are reduced to a lower oxidation state (e.g., $\text{Hf}^{4+} \rightarrow \text{Hf}^{3+}$, $\text{Ta}^{5+} \rightarrow \text{Ta}^{4+}$), the mechanism takes its name from this valence change [2]. Reversing the bias drives oxygen back. VCM is bipolar and is the family that includes HfO_x and TaO_x RRAM [1], [3].

The industry has converged on VCM, with HfO_2 and TaO_x as the two switching oxides that have crossed into foundry production [1], [5]. The reasons are made explicit later in the comparison with PCM and CBRAM (Section III-B), but in summary: HfO_2 is already qualified as the high- κ gate dielectric in modern HKMG MOSFETs, so deposition equipment and contamination protocols are already characterized in the fab, and the oxygen-vacancy filament is more thermally stable than the metallic filaments produced by ECM.

C. How VCMs Work

A VCM cell is a passive MIM stack: a transition-metal oxide a few nanometers thick is sandwiched between a top electrode and a bottom electrode (Fig. 3). When a voltage is applied, oxygen ions and oxygen vacancies migrate through the oxide and reorganize into a conductive filament that connects the two electrodes. Reversing or reducing the applied bias partially disconnects the filament, returning the cell to a high-resistance state [1], [3].

The full SET/RESET cycle is shown in Fig. 4. In the as-fabricated state the oxide is essentially an insulator (initial resistance state, IRS). A one-time soft dielectric breakdown step, called *forming* or *electroforming*, is required to nucleate the conductive filament [1]. During forming a voltage larger than the operational SET voltage is applied while the current is held to a compliance level I_{CC} , typically by a series transistor. The compliance current is essential because it limits the energy delivered into the dielectric and prevents an irreversible hard breakdown. After forming, the cell can be cycled at lower voltage between SET (HRS $\rightarrow$ LRS, soft breakdown that reconnects the filament) and RESET (LRS $\rightarrow$ HRS, partial rupture of the filament). In bipolar operation the SET and RESET pulses have opposite polarity; in unipolar operation they share polarity but differ in current level [1], [2].

In the LRS, conduction is approximately ohmic and dominated by the metallic-like sub-stoichiometric filament. In the HRS, conduction is more complex and depends on the material system: trap-assisted tunneling, Poole–Frenkel emission, and

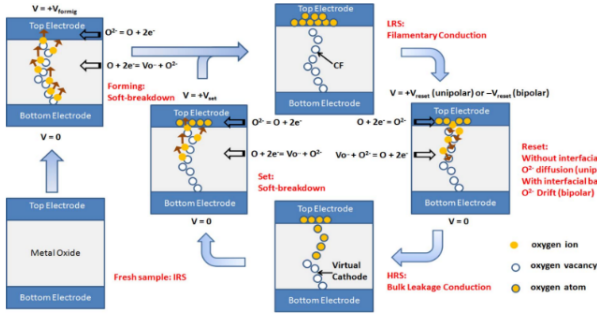


Fig. 4. Operation of a VCM cell. Starting from the as-fabricated insulating state, a forming pulse nucleates a conductive filament of oxygen vacancies. Subsequent SET pulses reconnect the filament to enter the LRS; RESET pulses partially rupture the filament to return to the HRS. Reproduced from [1].

Schottky emission at the oxide/electrode interface have all been reported [1]. The result is a non-trivial I - V shape that depends on electrode work function, oxide stoichiometry, and process conditions.

The hysteresis loop visible in measured RRAM I - V curves is a direct consequence of the cell's resistance being a function of internal state (filament configuration) rather than only of the instantaneous applied voltage. RESET partially ruptures the filament—it does *not* restore the oxide to its as-fabricated insulating state, and forming need not be repeated—so on the return sweep the cell carries the memory of the previous SET. This places RRAM in the broader memristor family proposed by Chua, in which the resistance is governed by an internal state variable that integrates the device history [3], [6].

III. WHY RRAM?

A. The eFlash Replacement Argument

The first and primary commercial driver for RRAM is the replacement of embedded flash at advanced nodes. Embedded flash does not scale below approximately 28 nm: the tunnel oxide cannot be thinned without sacrificing 10-year retention, and the high-voltage charge pumps required to program the cell add area and mask count that grow as a fraction of the die at advanced nodes [5]. RRAM avoids both problems. The MIM cell stack is on the order of 5–10 nm thick and does not rely on charge storage, so retention is governed by the thermal stability of the oxygen-vacancy distribution rather than by tunnel-oxide leakage. Programming voltages, while higher than core-logic V_{DD} , are an order of magnitude below those required for flash, which makes the on-die charge pump correspondingly smaller [1], [5].

A second advantage is back-end-of-line (BEOL) integration. The MIM stack is fabricated in the metal layers above the transistors, between two interconnect levels. The front-end logic flow is essentially unchanged, and the memory module adds only one or two extra masks [1]. This stands in sharp contrast to embedded flash, which requires its own front-end module. BEOL integration also opens a path toward stacking which planar charge-storage memories do not have.

B. Why VCM Specifically: Comparison with PCM and CBRAM

Of the three redox-based RRAM families, VCM is the only one that has reached significant production volume in embedded eFlash replacement. The reasons emerge from a head-to-head comparison with the two main competing mechanisms within the RRAM family, and with PCM as a closely related two-terminal NVM that shares much of the cell architecture but switches by a different physical mechanism.

VCM versus CBRAM/ECM. CBRAM cells use Ag or Cu as the active electrode, neither of which is welcome in a logic foundry: Cu is a deep-level recombination center in silicon and a contamination concern even with diffusion barriers, and Ag is not a standard CMOS material at all. Adopting CBRAM at a logic foundry therefore requires dedicated tooling and protocols that VCM does not [1], [2]. Beyond process compatibility, the physics also disfavors CBRAM for embedded NVM: the metallic filament is more sensitive to thermal activation than the oxygen-vacancy filament of a VCM cell, which complicates retention at the 105–150 °C operating temperatures required for automotive and IoT applications [2], [3]. CBRAM has nonetheless been commercialized in low-temperature niches (e.g., Adesto's CBRAM at 130 nm), but has not crossed into HKMG-class logic.

VCM versus PCM. PCM stores information in the crystalline (low-resistance) versus amorphous (high-resistance) state of a chalcogenide such as $\text{Ge}_2\text{Sb}_2\text{Te}_5$. A chalcogenide-based phase-change-like cell was the foundational technology of the Intel–Micron 3D XPoint memory (sold under the Optane brand), which shipped at meaningful volume from 2017 until its discontinuation in 2022; the precise switching mechanism was never publicly disclosed and is generally believed to be a phase-change-class storage element distinct from conventional $\text{Ge}_2\text{Sb}_2\text{Te}_5$ melt-quench PCM [5], and PCM is currently in production for embedded NVM at STMicroelectronics in a 28 nm FD-SOI automotive platform (the Stellar microcontroller family) [5]. PCM is therefore not a failed technology—it is a serious competitor, and notably ST's deployment on 28 nm FD-SOI places it directly in the same architectural niche that Weebit/GlobalFoundries occupy with TaO_x eRRAM on the 22FDX platform; eRRAM does not have FD-SOI to itself. The reasons VCM has nonetheless taken the larger share of embedded RRAM-class deployments are primarily about current and thermal envelope:

- PCM RESET (amorphization) requires heating the chalcogenide above its melting temperature ($T_m \approx 600$ °C for $\text{Ge}_2\text{Sb}_2\text{Te}_5$) followed by rapid quenching, while SET (crystallization) holds the cell between the glass transition T_g and T_m to allow nucleation and growth [9]. The melt-quench requirement is what drives RESET currents to ~ 100 μA or more in scaled cells. The resulting select-FET width is several times larger than the corresponding VCM 1T1R, eroding the cell-area win [1], [5].
- Heat from a PCM cell can disturb its neighbors, particularly at small pitch, which limits density. VCM cells dissipate orders of magnitude less power per switching event and have no such thermal crosstalk pathway [3].

- HfO_2 and TaO_x , the VCM workhorse oxides, are already qualified materials in HKMG flows; chalcogenides are not.

PCM remains advantageous when very tight resistance distributions or genuinely binary states are required (notably, 3D XPoint exploited tight chalcogenide-cell distributions to enable selectorless 3D crosspoint operation). For embedded eFlash replacement at moderate write rates, VCM is generally preferred.

VCM versus STT-MRAM. STT-MRAM is not within the RRAM family but is the dominant emerging eNVM at FinFET-class nodes and therefore the most relevant external comparison. STT-MRAM has substantially higher endurance (10^{14} vs. 10^6 – 10^9 cycles), no forming step, and deterministic switching. RRAM wins on cell area (filamentary cells are smaller than the magnetic tunnel junction stack with its ferromagnetic reference layer), on mask adder, and on continuously programmable conductance for analog applications. The result is that the two technologies coexist in the embedded NVM market: MRAM at sub-22 nm FinFET nodes where endurance and density-per-cache-line dominate, RRAM at planar/FD-SOI nodes where mask count, area, and write-rare workloads dominate [5].

C. The Emerging NVM Landscape

By the mid-2020s embedded RRAM is a foundry-process option at multiple companies. Fig. 5 reproduces the industry timeline assembled by Hellenbrand *et al.* for FRAM, MRAM, PCRAM, and RRAM from 2017 through 2026 [5]. RRAM and MRAM together account for the majority of recent announcements, and their respective domains in the industry chart are visible at a glance: eMRAM populates the FinFET-class nodes (16 nm, 14 nm, 8 nm) at TSMC, Samsung, and GlobalFoundries, while eRRAM clusters at the planar bulk and FD-SOI nodes (22–40 nm). PCM is sparser in the chart but is in active production at one major automotive supplier. The clear practical implication is that the eMRAM ecosystem is currently more mature than the eRRAM ecosystem at FinFET-class nodes, and eRRAM has not yet caught up; whether it does is one of the open questions for the next several process generations.

The headline RRAM deployments are:

- *TSMC.* HfO_2 -based embedded RRAM at 40 nm [7], 28 nm, and 22 nm planar bulk, with a 12 nm FinFET demonstration presented at IEDM 2023 [5]. The 40 nm macro of Chou *et al.* reports a $256\text{K} \times 44$ array with on-chip current limiting to enable low-voltage SET and RESET, illustrating the practical interplay between cell sizing and peripheral circuit design.
- *Intel.* A 22 nm FinFET RRAM cell using a TaO_x -based switching oxide, integrated into the 22FFL low-power FinFET platform; reported endurance is $\sim 10^4$ cycles, which is a representative production-grade rather than laboratory number [8].
- *Weebit Nano + GlobalFoundries / SkyWater.* TaO_x ReRAM IP integrated into the 22FDX FD-SOI platform and the SkyWater 130 nm bulk platform [5].

- *IBM.* Investigating both TaO_x and HfO_2 stacks for analog in-memory compute, exploiting the continuous conductance levels available in VCM cells [5], [6].

The applications are nearly exclusively embedded: firmware code, calibration constants, AI inference weights, and secure keys in microcontrollers, automotive ICs, IoT system-on-chips, and secure elements [5]. What unifies these workloads is that writes are infrequent (often write-once at provisioning), retention requirements are stringent, and security or latency forbids using an external SPI flash chip. RRAM has not displaced DRAM in working memory and is not a credible SRAM replacement: SRAM operates at sub-1 V V_{DD} and at write rates that would exceed RRAM endurance within hours of operation, while RRAM SET requires 1.5–2 V at the bitline and forming requires 3–4 V.

D. Position in the Memory Hierarchy

A useful way to see RRAM's niche is to place it directly against the dominant memory technologies. Table I summarizes the comparison.

The pattern is that RRAM is an embedded non-volatile technology with cell area between DRAM and SRAM, fast read, medium write, per-bit programmability, and a write voltage that requires only a modest on-die boost. Its endurance in production (10^4 – 10^6 cycles for HfO_x macros, with TaO_x reaching up to 10^{12} in lab demonstrators) is comparable to or moderately better than flash but several orders worse than DRAM, which is why RRAM has not displaced DRAM in working memory. Its niche is precisely the embedded NVM role formerly held by NOR flash, with substantially better scalability and a credible path to denser arrays.

IV. RRAM IN CMOS: 1T1R INTEGRATION

A. Competing Architectures: 1T1R and the Crossbar

At the array level there are two competing architectures, illustrated in Fig. 6. In the 1T1R architecture, every cell carries its own select MOSFET, fabricated in the front-end logic flow, with the RRAM stack placed in the BEOL above it [4]. In the crossbar architecture, the RRAM cells sit at the intersections of orthogonal bitlines and wordlines, with no per-cell select element; addressing relies on biasing the selected lines and the unselected lines at different voltages.

The two architectures use very different bias schemes, summarized in Table II. In 1T1R, the select MOSFET clamps the cell current and isolates unselected rows, so the bias scheme is straightforward: drive the bitline (BL) to the read or write voltage, hold the source line (SL) at ground, and use the wordline (WL) to gate the selected MOSFET into triode (read) or saturation (write). In the crossbar, no transistor is available, so a half-select scheme is used: the selected wordline is biased at $+V_w/2$ and the selected bitline at $-V_w/2$ (or the read variant at $\pm V_r/2$), so that only the targeted intersection sees the full programming voltage. Half-selected cells still see $V_w/2$ across them, however, and the resulting sneak current

$$I_{\text{sneak}} = \frac{(N-1)V_r}{2R_{\text{LRS}}} \quad (1)$$

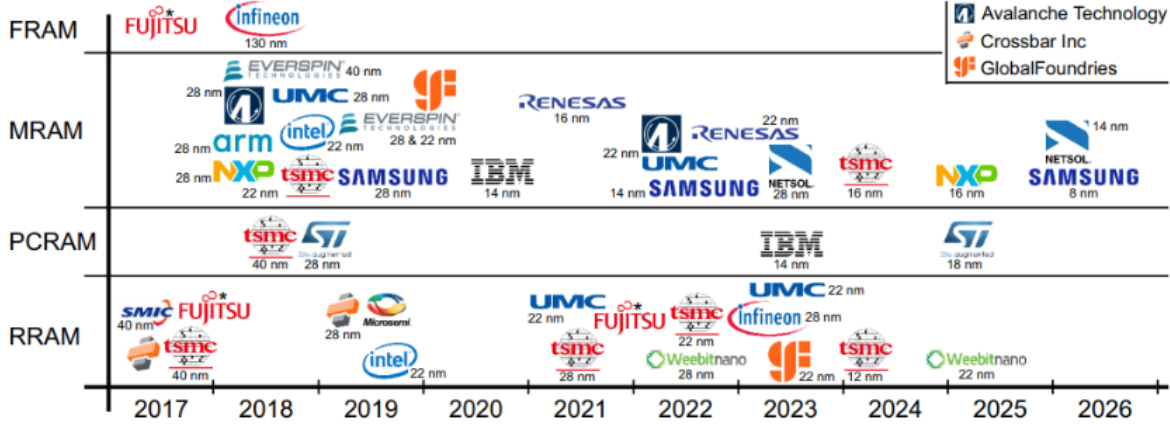


Fig. 5. Industry roadmap for emerging non-volatile memories from 2017 through 2026. eMRAM and eRRAM together account for most of the recent announcements; eMRAM populates the FinFET-class advanced nodes while eRRAM clusters at planar bulk and FD-SOI. Reproduced from [5].

TABLE I
COMPARISON OF RRAM AND CONVENTIONAL MEMORY TECHNOLOGIES

	SRAM	DRAM	Flash NAND	Flash NOR	RRAM
Volatile?	Volatile	Volatile	Non-volatile	Non-volatile	Non-volatile
Cell size	$\sim 50 F^2$	$\sim 6 F^2$	$\sim 4 F^2$	$\sim 10 F^2$	$\sim 6\text{--}12 F^2$
Read speed	Fast	Fast	Slow	Fast	Fast
Write speed	Fast	Fast	Slow	Slow	Medium
Endurance	Unlimited	$\sim 10^{15}$	$10^4\text{--}10^5$	$10^4\text{--}10^5$	$10^4\text{--}10^6$ prod.; 10^{12} lab
Write voltage	V_{DD}	V_{DD}	~ 20 V	~ 20 V	$V_{DD} + \text{boost}$ ~ 2 V
Granularity	per-bit	per-bit	block-erase	block-erase	per-bit
On-chip use	Yes	No	No	Yes	Yes

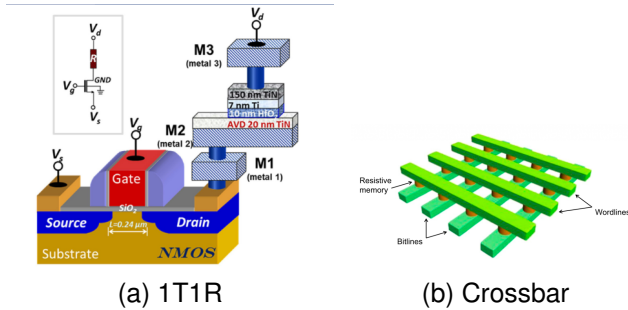


Fig. 6. Competing RRAM array architectures. (a) 1T1R: a CMOS NMOS in the front-end with the $\text{HfO}_2/\text{Ti}/\text{TiN}$ MIM stack between metals 2 and 3. Reproduced from [4]. (b) Passive crossbar with bitlines, wordlines, and resistive memory at every intersection. Reproduced from [3].

limits the maximum array fan-out N at which the read margin remains acceptable [3]. The crossbar promises a theoretical $4F^2$ cell footprint and a path to true three-dimensional stacking, but only works in production if the sneak current is suppressed by an effective two-terminal selector at every node. No such selector has yet entered foundry production for embedded RRAM, which is why every commercial RRAM macro shipping today is 1T1R.

B. Why 1T1R: Mapping RRAM Requirements to Transistor Design

The 1T1R cell is preferred in production because the MOSFET solves three RRAM problems that the crossbar cannot solve at the array level. Each requirement maps directly to a transistor design knob, summarized in Table III.

The first row is the most quantitative. During SET (and during forming, where the requirement is even tighter), the RRAM cell must be current-limited or it will undergo hard breakdown. The select MOSFET is biased into saturation to act as a current source, and is sized so that its saturation current matches the desired compliance I_{CC} :

$$I_D = I_{CC} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2. \quad (2)$$

The compliance current itself is a design choice that trades cell endurance against R_{LRS} : lower I_{CC} produces a thinner filament with higher R_{LRS} but better endurance and lower switching energy [3]. The trade-off is shown in Fig. 7, which plots LRS resistance and RESET current against I_{CC} for several oxide systems. Both quantities scale with I_{CC} over more than two decades, which is what allows multilevel and analog operation of the cell. From the MOSFET side, hitting a target I_{CC} becomes a problem of W/L sizing combined with V_{GS} control through the wordline.

TABLE II
BIAS SCHEMES FOR 1T1R AND CROSSBAR ARRAYS

Operation	1T1R	Crossbar
READ	BL: 0.3–0.4 V; SL: 0 V; WL: V_{DD} . MOSFET in triode; sense I_{cell} vs. I_{ref} ; $R_{HRS}/R_{LRS} \sim 10\text{--}1000\times$ in HfO_2 .	WL: $-V_r/2$, BL: $+V_r/2$. Sneak current $I_{sneak} = (N-1)V_r/(2R_{LRS})$ limits array fan-out.
SET	BL: 2–2.5 V; SL: 0 V; WL: V_{DD} . MOSFET in saturation, used to limit sudden current jump; $R_{LRS} \approx V_c/I_{CC}$.	WL: $+V_w/2$, BL: $-V_w/2$. Cannot limit sudden current jump when channel conducts; $N-1$ half-selected cells.
RESET	SL: 1.5 V; BL: 0 V; WL: V_{DD} . MOSFET in saturation (reverse polarity).	BL: $+V_w/2$, WL: $-V_w/2$. $N-1$ half-selected cells on each bias line.

Compiled from [3], [4].

TABLE III
RRAM REQUIREMENTS MAPPED TO SELECT-FET DESIGN KNOBS

RRAM requirement	Transistor design knob
Limit I_{CC} for SET (and forming)	Operate in saturation with built-in headroom; size W/L so that $I_D = I_{CC} = \frac{1}{2}\mu_n C_{ox} (W/L)(V_{GS} - V_T)^2$.
Suppress sneak leakage from unselected cells	Limit DIBL; minimize subthreshold swing $SS = (kT/q) \ln 10 (1 + C_d/C_{ox})$.
Ensure clean read margin	Limit R_{on} of the transistor so that $R_{LRS} \gg R_{on}$.

and the BEOL. The on-resistance of the select MOSFET, $R_{on} = L/[\mu_n C_{ox} W (V_{GS} - V_T)]$, must remain much smaller than R_{LRS} , otherwise the read voltage divides between the cell and the transistor and the HRS/LRS distinction collapses.

C. A Worked Example: Niu 2016 Planar Process

Niu *et al.* provide an explicit worked example of the sizing problem on a 0.25 μm planar CMOS line [4]. The cell is a $\text{HfO}_2/\text{Ti}/\text{TiN}$ MIM stack placed between metal 2 and metal 3 above an NMOS in the front-end. Starting from the series-circuit constraint

$$V_{BL} = V_{RRAM} + V_{DS}, \quad (3)$$

and requiring the MOSFET to remain in saturation during forming and SET ($V_{DS} > V_{GS} - V_T$), the design constraint becomes $V_{BL} - V_{SET} > V_{GS} - V_T$. Choosing $W = 1.14 \mu\text{m}$ and $L = 0.24 \mu\text{m}$ at $V_g = 0.9 \text{ V}$ sets the saturation current to $I_{CC} = 10 \mu\text{A}$ by Eq. (2). The resulting LRS resistance is $R_{LRS} = V_c/I_{CC} = (0.4 \text{ V})/(10 \mu\text{A}) = 40 \text{ k}\Omega$ (with $V_c \approx 0.4 \text{ V}$ for HfO_2), while the transistor on-resistance from the saturation model is $R_{on} \approx 200 \Omega$, comfortably below the 40 k Ω LRS as required by the read-margin constraint. Forming requires a higher voltage of 3.2–4.0 V on the bitline; the post-forming SET voltage falls to 0.8–1.1 V and RESET to 1.0–1.5 V [4].

The measured I – V curves of this 1T1R cell, for two different ALD deposition temperatures, are shown in Fig. 8. Higher deposition temperature reduces residual carbon impurities from the metal-organic precursor and stabilizes the resistance distributions, lowering both V_{SET} and V_{RESET} and tightening cycle-to-cycle variability [4]. The case study makes the broader point that VCM RRAM electrical metrics are inseparable from process choices on the underlying materials stack.

V. RELIABILITY AND DISTURB MECHANISMS

A. Endurance and the Fresh-Oxide Question

A natural question for any non-volatile memory is whether each RESET fully restores the cell to its as-fabricated state. For RRAM the answer is no, and the distinction matters. Forming, which performs the initial soft dielectric breakdown, happens once. The first SET reuses the path opened by forming; every subsequent SET and RESET reorganizes oxygen vacancies along (or near) that same path. The oxide therefore retains a memory of every cycle: oxygen ions accumulate at electrode

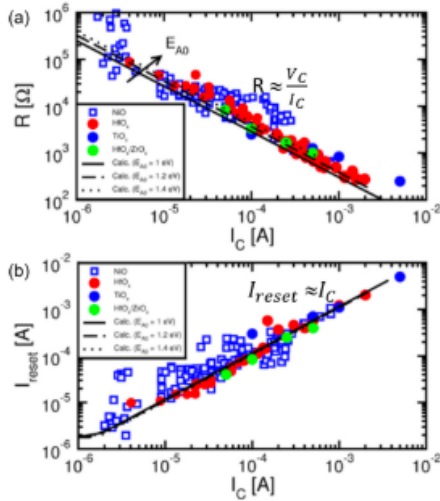


Fig. 7. Dependence of the LRS resistance (top) and the RESET current (bottom) on the compliance current I_{CC} across several VCM oxide systems. $R_{LRS} \propto V_c/I_{CC}$ and $I_{reset} \approx I_{CC}$. Reproduced from [3].

The second row constrains the off-state of the select FET. In a large array, every unselected cell sees a small bias and contributes a subthreshold leakage current; these add up across the array and degrade the read margin. The relevant figure of merit is the subthreshold swing $SS = (kT/q) \ln 10 (1 + C_d/C_{ox})$, which sets how steeply I_D falls below V_T . A steeper SS means more cells can share a wordline before the cumulative leakage corrupts a read; this is one of the main reasons FinFETs and FD-SOI MOSFETs (with smaller C_d/C_{ox}) are attractive hosts for RRAM at advanced nodes. DIBL is also constrained because it shifts V_T at high V_{DS} , which would alter the saturation I_D used to set I_{CC} .

The third row is a co-optimization between the MOSFET

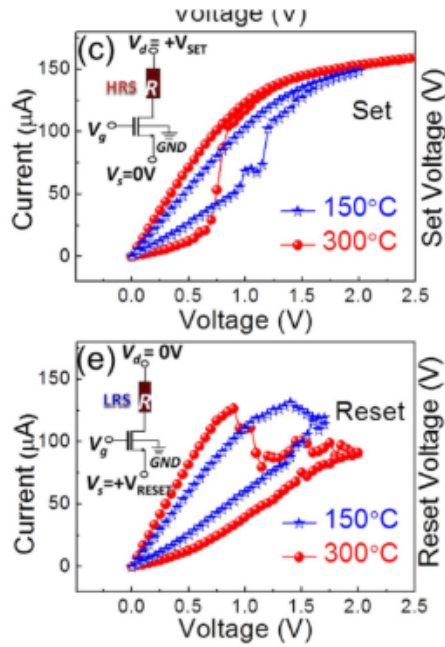


Fig. 8. Measured I - V characteristics of the integrated 1T1R HfO_2 cell from [4], comparing two atomic layer deposition temperatures (150 °C in blue and 300 °C in red). Top: SET; Bottom: RESET. The 300 °C deposition produces tighter resistance distributions and lower switching voltages. Reproduced from [4].

interfaces, lattice strain builds up, and a small fraction of defects become permanently fixed in positions that no SET pulse can move [3]. Endurance, defined as the number of cycles for which a programmable read margin is maintained, is the time to failure of this gradually wearing path.

In practice, production HfO_x cells reach 10^4 – 10^6 cycles, and laboratory TaO_x cells have demonstrated up to 10^{12} cycles in single devices, suggesting that the ceiling is materials- and stack-engineering limited rather than fundamental [3], [5]. Two failure modes dominate. In one, the LRS resistance drifts upward over cycles because the filament cannot be fully reformed; in the other, the HRS resistance drifts downward because residual conductive paths persist after RESET. Either way, the read window between LRS and HRS narrows until a comparator can no longer discriminate the two states. Endurance and retention are tightly coupled to the same process variables (oxide stoichiometry, electrode work function, getter layer thickness), which is why they are usually optimized as a single triple alongside switching uniformity [3], [4].

B. Retention and Read Disturb

Retention is a ten-year claim about the thermal stability of the programmed resistance distribution. Empirically, LRS retention is governed by Arrhenius-activated dissolution of the filament, with extracted activation energies in the 1.0–1.5 eV range for HfO_x [3]. The HRS distribution is also non-stationary in time: even small read voltages slowly redistribute oxygen vacancies and broaden the resistance distribution, producing what is conventionally called read disturb.

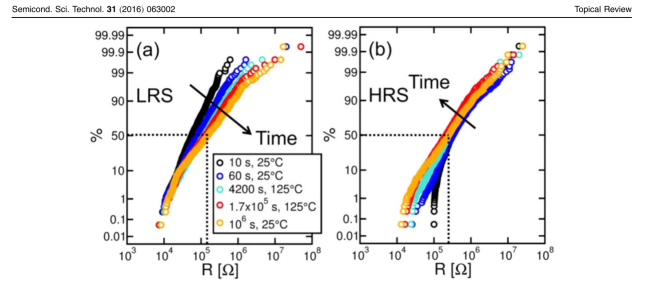


Figure 20. Statistical study of retention for LRS and HRS. Both LRS (a) and HRS distributions (b) indicate broadening with time. Even if the value of HRS resistance is forced above $10^3 \Omega$, the distribution soon extends to lower resistance values due to noise. Copyright (2015) IEEE. Reprinted, with permission, from [109].

Fig. 9. Statistical study of retention for the LRS (a) and HRS (b) distributions in a HfO_2 RRAM cell at multiple temperatures and elapsed times. Both distributions broaden with time, narrowing the read window; the HRS tail in particular drifts toward lower resistance and is the dominant read-disturb signature. Reproduced from [3].

Fig. 9 shows the dominant disturb signature in a HfO_2 1T1R cell: the LRS and HRS distributions broaden with time, with the broadening accelerated at elevated temperature [3]. The HRS tail in particular tends to drift toward lower resistance, narrowing the read window from above. This is the underlying reason that retention temperature and array bake tests are central to RRAM qualification: a cell that is comfortably in HRS at 25 °C may have drifted into the read-uncertain region at 125 °C after several years.

In the 1T1R architecture, write disturb on neighboring cells is small because the select FET isolates each row, and read disturb at the bit level is well-controlled because the read voltage (0.3–0.4 V) is far below the SET threshold. The dominant disturb in 1T1R is the cumulative time-and-temperature drift of the resistance distributions described above. The crossbar architecture additionally suffers from program disturb: the half-selected cells along the selected wordline and bitline see $V_w/2$ during every write, and over many writes this drives the same Arrhenius drift that is otherwise gradual. This is one more reason that crossbars require a strongly nonlinear selector to stop charge from flowing through half-selected cells.

C. Voltage Envelope and Charge-Pump Implications

A practical question for any embedded NVM is whether the program voltages can be generated on-chip from the core V_{DD} , or whether an off-chip supply is required. Forming a HfO_x cell needs 3.2–4.0 V at the bitline; SET and RESET need 1.5–2 V; read needs 0.3–0.4 V [4]. At a per-cell SET current of 10–50 μA , even a kilobit-scale program operation draws on the order of milliamps. Both the voltage and the current are well within the envelope that on-die switched-capacitor charge pumps can deliver, and every shipping embedded RRAM macro generates these voltages internally; no off-chip high-voltage supply is required. This is in sharp contrast to flash, which needs 15–20 V at the cell and therefore a much larger on-die pump network [1], [5]. The charge-pump area savings are part of why RRAM's mask adder ends up smaller than flash even though the cell stack itself is similar in complexity.

TABLE IV
COMPARISON OF MOSFET ARCHITECTURES HOSTING RRAM

Parameter	Planar bulk	FD-SOI	FinFET
V_{DD}	1.0 V	0.9–1.0 V	0.8–0.9 V
I_{CC} control	$W/L + V_g$	$W/L + V_g + \text{back-gate}$	$N_{fin} + V_g$
SS (mV/dec)	80–100	65–75	~65
Min. cell area	$0.06 \mu\text{m}^2$	$0.066 \mu\text{m}^2$	$0.022 \mu\text{m}^2$
Forming headroom	OK	charge pump may be needed	charge pump required
Production eRRAM	TSMC 28 nm	STMicro (22FDX)	TSMC 12 nm

Adapted from [3] and [5].

VI. TECHNOLOGY-DEPENDENT CHALLENGES: PLANAR, FD-SOI, AND FINFET

The same RRAM cell is integrated on top of three different MOSFET architectures in production today. Because the select-FET problem changes character with each architecture, the achievable cell density, I_{CC} control granularity, and forming headroom all change as well. Table IV summarizes the comparison.

Planar bulk (250 nm down to 28 nm). The starting point. Planar bulk MOSFETs offer continuous sizing of W and L , which is the most flexible knob for tuning I_{CC} . The penalty is the worst subthreshold slope of the three architectures (80–100 mV/dec), which limits the maximum array fan-out. TSMC’s 40 nm and 28 nm embedded RRAM products are on planar bulk [5], [7], as is the Niu 2016 case study above.

FD-SOI (28FDSOI Samsung, 22FDX GlobalFoundries). The fully-depleted silicon-on-insulator transistor adds a back-gate that can be biased independently of the front gate. In the 1T1R cell this back-gate provides per-die threshold trim of the select FET, which translates into per-die I_{CC} trim—useful for compensating both the process variation in the FET and the much larger device-to-device variation in the RRAM cell itself. The subthreshold slope improves to 65–75 mV/dec because the buried oxide effectively eliminates the body capacitance term C_d . The trade-off is a slightly larger minimum cell area and a marginal forming headroom, since V_{DD} is similar to or below the planar value while the forming voltage at the cell remains near 3 V. Weebit Nano’s TaO_x ReRAM IP has been integrated into the 22FDX FD-SOI platform [5].

FinFET (16 nm and below). The FinFET delivers the steepest subthreshold slope of any production architecture (~65 mV/dec), which raises the achievable array fan-out, and provides higher drive current per footprint, which helps deliver I_{CC} in a small cell. The cell area is the smallest of the three, at roughly $0.022 \mu\text{m}^2$. The trade-off is geometric: W is no longer continuous but quantized into integer fins. Sizing for a target I_{CC} may force rounding to the next fin count, which shows up as a granular I_{CC} ladder rather than a smooth one. In addition, the lower V_{DD} (0.8–0.9 V) leaves less headroom to deliver the forming voltage at the cell, which makes a dedicated charge pump effectively mandatory. TSMC presented an embedded RRAM macro on its 12 nm FinFET technology at IEDM 2023 [5].

The dependence of the cell electrical characteristics on I_{CC} , common to all three architectures, is shown in Fig. 10. The

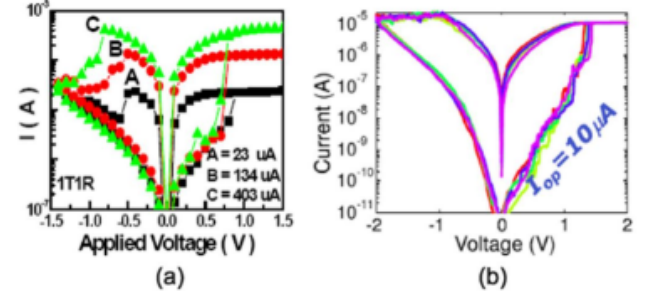


Fig. 10. Measured I - V characteristics of an integrated 1T1R HfO_x cell at three compliance current levels (a), and the resulting LRS/HRS distributions versus operating compliance (b). The compliance current is set by the select MOSFET. Reproduced from [3].

select MOSFET sets the compliance, and the LRS resistance, RESET current, and read margin all track it. The architectural choice therefore matters not only for area and array size but for the achievable granularity of the I_{CC} knob—continuous on planar bulk, continuously trimmable on FD-SOI, and quantized on FinFET.

VII. CONCLUSION AND OUTLOOK

RRAM is one of three serious contenders, alongside STT-MRAM and PCM, to replace embedded flash at logic nodes that have outgrown the floating-gate stack. Within RRAM, the industry has converged on the valence-change mechanism using HfO₂ or TaO_x as the switching oxide, for reasons that are concrete rather than abstract: the oxides are already qualified materials in HKMG flows, the oxygen-vacancy filament is more thermally stable than the metallic filaments of CBRAM, and the switching current is an order of magnitude smaller than the melt-quench currents of PCM. Each of those is a process-level or device-level fact that competes with the shared advantages of any MIM cell—small footprint, two-terminal simplicity, BEOL placement above the transistors.

The MOSFET-modeling content of the technology is concentrated in the select FET. Three RRAM electrical requirements—compliance-current limiting during SET and forming, suppression of sneak leakage from unselected cells, and a clean read margin—map directly onto three transistor design knobs: W/L sizing in the saturation current model, low subthreshold slope and DIBL, and low R_{on} . The Niu 2016 case study on a $0.25 \mu\text{m}$ planar process makes this mapping explicit. Moving to FD-SOI adds a back-gate that gives per-die I_{CC} trim; moving to FinFET drops the cell area and the subthreshold slope but quantizes the sizing knob and tightens the forming headroom. Each architectural step on the MOSFET roadmap reshapes the select-FET problem rather than eliminating it, which is one reason RRAM, MRAM, and PCM each face a different practical integration story at advanced nodes despite sharing a common 1T1R cell template.

The most active forward-looking direction for RRAM is not as a memory replacement at all but as the storage element of an analog matrix-vector multiplier for AI inference. The continuous, programmable conductance of a VCM cell—accessible by varying the compliance current during SET—turns the

array into a physical implementation of Ohm's law and Kirchhoff's current law, performing the dot products at the heart of neural-network inference directly inside the memory and bypassing the von Neumann bottleneck [6]. Several foundries (TSMC, Samsung, IBM) and academic groups have published RRAM-based in-memory compute demonstrations using this approach [5], [6]; representatively, Yao *et al.* demonstrated a fully-integrated memristor convolutional neural network on a CMOS-compatible RRAM array, achieving accuracy comparable to a software-trained baseline on standard image-recognition benchmarks [10]. If RRAM does win a role beyond embedded NVM, this is the most plausible one—not as DRAM or SRAM replacement, but as the compute fabric for workloads where the memory and the multiplier are physically the same device.

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